

K-11

RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR

COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN HIGHER
NATIONAL DIPLOMA EXAM (HND)

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

MINISTRY OF HIGHER EDUCATION

NATIONAL COMMISSION FOR THE ORGANISATION OF
HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM

National Exam of Higher National Diploma – 2021 Session

Spécialty/option : CSN and SWE

Paper : Digital Electronics

Duration : 4 hours

Credits : 7

SECTION A: NUMBER SYSTEMS AND CODES (20 MARKS)

MCQs, four (04) answers are proposed circle the correct answer.

- 1) The hexadecimal number for $\square 95.5 \square 10$ is
 - A. $\square \square \square 5F.8 \text{ 16}$
 - B. $\square \square \square 9A.B \text{ 16}$
 - C. $\square \square \square 2E.F \text{ 16}$
 - D. $\square \square \square 5A.4 \text{ 16}$
- 2) Digital system is usually operated on.....system
 - A. Binary
 - B. decimal
 - C. Octal
 - D. hexadecimal
- 3) The binary system uses powers of.....for position values
 - A. 2B. 10
 - C. 8D. 16
- 4) After counting 0,1,10,11 the next binary is
 - A. 12
 - B. 100C. 101
 - D. 110
- 5) The number 1000_2 is equivalent to decimal number
 - A. One thousand
 - B. eight
 - C. Four
 - D. sixteen
- 6) The binary addition $1+1+1$ gives
 - A. 111
 - B. 10
 - C. 110
 - D. 11
- 7) The results of binary subtraction $(100-011)$ is
 - A. 111
 - B. 111
 - C. 011
 - D. 001
- 8) The 2 's complement of 1000_2 is
 - A. 0111
 - B. 0101
 - C. 1000
 - D. 0001
- 9) The code where all successive numbers differ from their preceding number by single bit is
 - A. Binary code
 - B. BCD
 - C. Excess -3
 - D. Gray
- 10) Octal coding involves grouping the bit's in
 - A. 5's
 - B. 7's
 - C. 4's
 - D. 3's

- 11) In Excess-3 code each coded number is..... than in BCD code.
 A. four larger B. three smaller C. three larger D. much larger.
- 12) which numbering system uses numbers and letters as symbols
 A. decimal B. binary C. Octal D. hexadecimal
- 13) To convert a whole decimal number into a hexadecimal equivalent, one should divide the decimal value by.....
 A. 2 B. 8 C. 10 D. 16
- 14) The number 12_8 is equivalent to decimal
 A. 12 B. 20 C. 10 D. 4
- 15) The result of binary multiplication $111_2 \times 10_2$ gives
 A. 1101 B. 0110 C. 1001 D. 1110
- 16) A device which converts BCD to Seven Segment is called
 A. Encoder B. Decoder C. Multiplexer D. Demultiplexer
- 17) 1's complement representation of decimal number of -17 by using 8 bit representation is
 A. 1110 1110 B. 1101 1101 C. 1100 1100 D. 0001 0001
- 18) The excess 3 code of decimal number 26 is
 A. 0100 1001 B. 01011001 C. 1000 1001 D. 01001101
- 19) How many AND gates are required to realize $Y = CD + EF + G$
 A. 4 B. 5 C. 3 D. 2
- 20) How many select lines will a 16 to 1 multiplexer will have
 A. 4 B. 3 C. 5 D. 1

SECTION B: COMPUTER FUNDAMENTALS (80 MARKS)

Question 1 : Hardware (20 Marks)

- Give 3 basic functionalities of a computer (3 marks)
- Which computer component contains all the circuitry necessary for other components or devices to communicate with one another? (1 mark)
- what is the role of the chipsets? (2 marks)
- The computer memory can be classified into primary and secondary memory. Give two examples of each. (2 marks)
- Give 4 differences between the RAM and a HDD (4 marks)
- List 3 computer input and 3 computer output devices (3 marks)
- Explain briefly the steps necessary for a computer to boot (4 marks)
- What is a Bus? (1 mark)

Question 2 :Network and Mobile Devices (40 Marks)

- Classify networks in terms of their geographic sizes and for each of the networks, describe them briefly (6 marks)
- What do you understand by network topology? (1 mark)

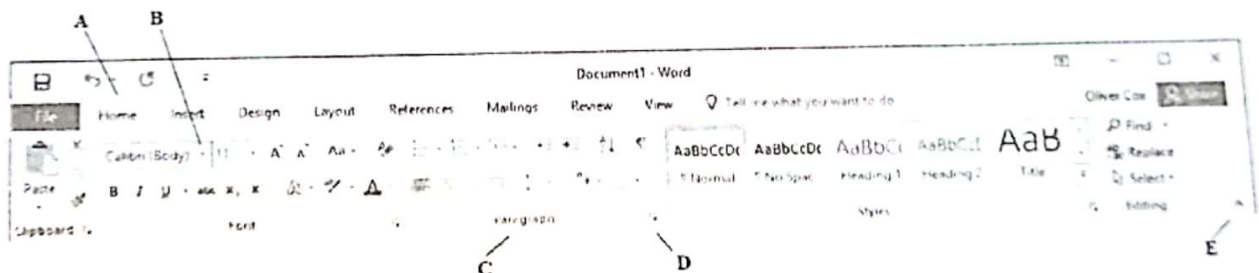
- c) Briefly explain the network topologies given below with the aid of diagrams
- Bus topology
 - Star topology
 - Ring topology
 - Meshtopology. (4x2= 8 marks)

The OSI model breaks the various aspects of a computer network into seven distinct layers.

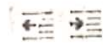
- d) What is the full meaning of OSI? (1 mark)
- e) Give the seven layers of the OSI model and describe them, in either ascending or descending order. (14 marks)
- f) The following are network connectivity devices briefly describe their role
- Modem
 - Hub
 - Switches
 - Routers (4x2= 8 marks)
- g) What is GSM? (2 marks)

Question 3 :Microsoft Word, Excel and PowerPoint (20 Marks)

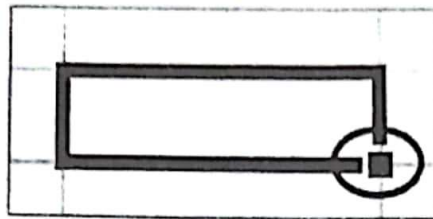
Consider the figure shown below.



- How do we call this figure? (1 mark)
- Name the part A through E? (3 marks)
- What is the difference between save as? (2 marks)
- What is the name and use of the symbols shown in the image below? (2 marks)



- What name is given to the figure in the circle below? (1 mark)



- f) Copy and complete the tables below when the symbol above is pulled downward.(3 marks)

Text

123

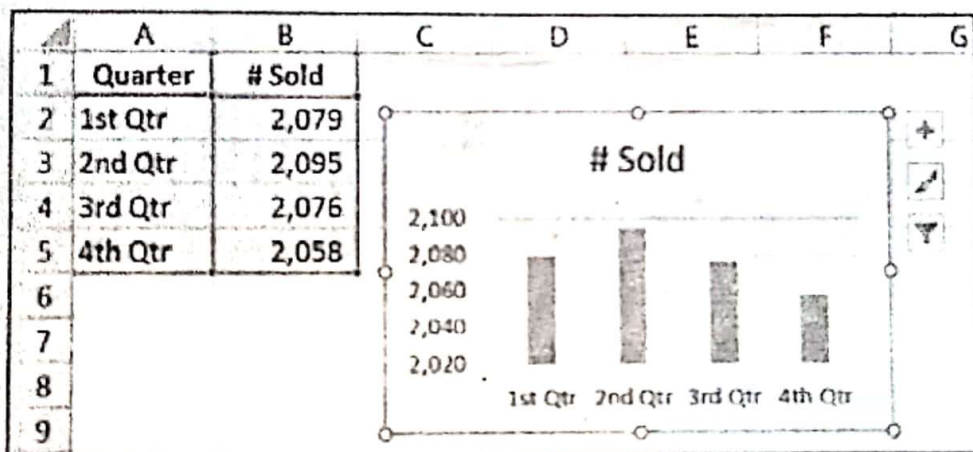
Exam 1

2/1/02

February

Friday

- g) Consider the image below, what is the procedure necessary to obtain the chart shown on the diagram(2 marks)



- h) What is the use of the function autoSum? (2 marks)
 i) What is the importance of a PowerPoint? (2 marks)
 j) What are the extensions of a word, excel and PowerPoint file? (2 marks)